

Ex No :	Function Component
Date :	

AIM:

To implement a application on react using Function Component for BusTracker, Pet Adoption and Real-Time Auction platform.

PROGRAM:**BusTracker.jsx**

```
import React, { Component } from "react";
import "../BusTracker.css";
class BusTracker extends Component {
  constructor(props) {
    super(props);
    this.state = {
      eta: props.eta
    };
  }
  refreshETA = () => {
    const newETA = Math.floor(Math.random() * 15) + 1;
    this.setState({ eta: newETA });
  };
  getETAColor = () => {
    if (this.state.eta < 5) {
      return "green";
    } else if (this.state.eta <= 10) {
      return "yellow";
    } else {
      return "red";
    }
  }
}
```

```
};  
  
render() {  
  const { busNumber, route } = this.props;  
  const { eta } = this.state;  
  return (  
    <div className="bus-card">  
      <div className="bus-icon"><img alt="bus icon" data-bbox="400 260 420 275"/></div>  
      <h1>Transport </h1>  
      <div className="bus-number">  
        Bus {busNumber}  
      </div>  
      <p className="route">  
        <img alt="location pin icon" data-bbox="175 430 195 445"/> Route: {route}  
      </p>  
      <div className={`eta ${this.getETAColor()}`}>  
        {eta === 0 ? "Arriving Now" : `${eta} minutes`}  
      </div>  
      <p className="status">  
        Estimated Arrival Time  
      </p>  
      <button onClick={this.refreshETA}>  
        <img alt="refresh icon" data-bbox="175 690 195 705"/> Refresh ETA  
      </button>  
    </div>  
  );  
}  
}  
  
export default BusTracker;
```

App.jsx:

```
import BusTracker from "../BusTracker";

function App() {

  return (

    <BusTracker

      busNumber="101"

      route="Tirunelveli → Palyamkottai"

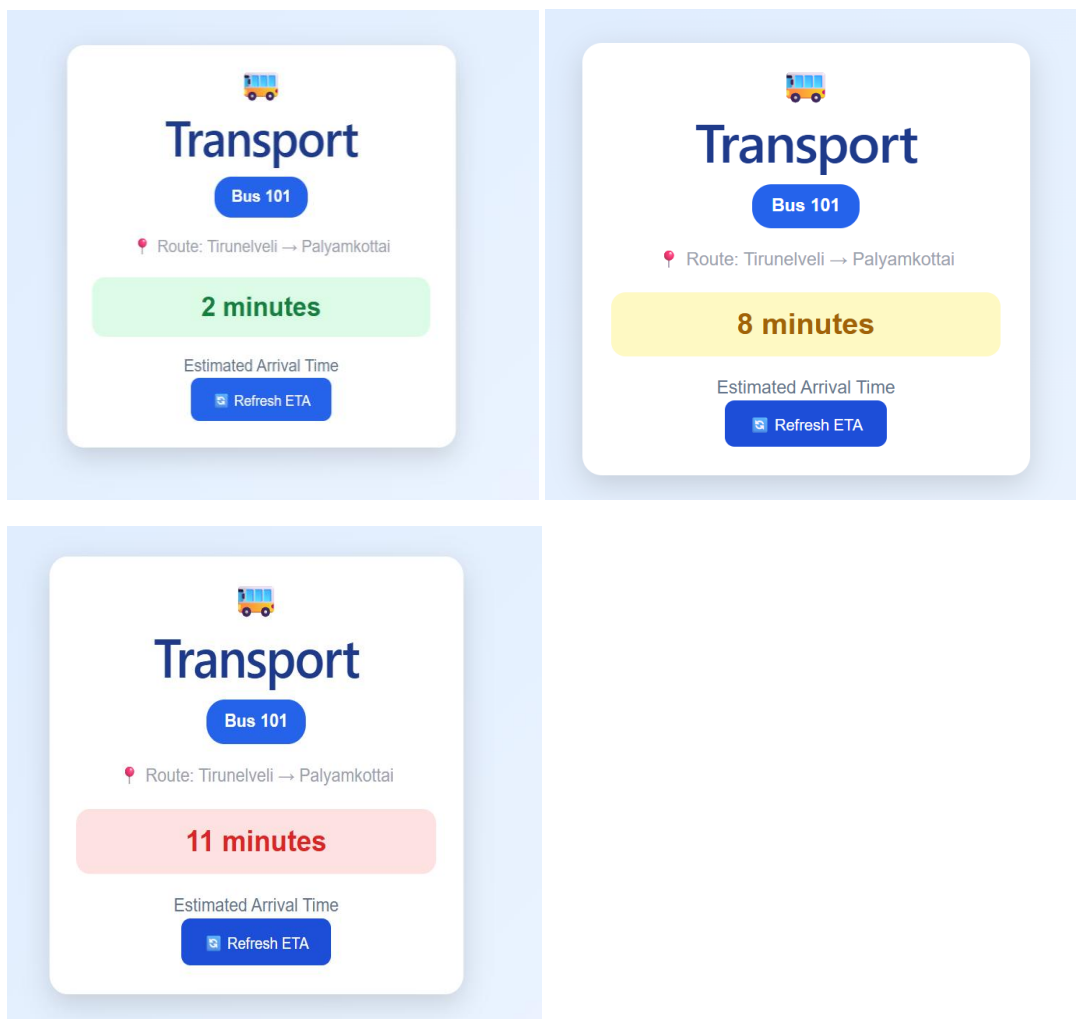
      eta={7}

    />

  );

}

export default App;
```

OUTPUT:

2. Pet Adoption Platform

Scenario: Build a platform for showcasing pets available for adoption.

Tasks:

1. Create a PetCard class component that:
 - Displays the pet's name, age, species, and a brief description as props.
 - Adds a badge for "Young" pets if the age is less than 2 years.
2. Add a "Toggle Details" button that shows or hides the full description of the pet.
3. If the species is "dog", display a 🐶 icon; if "cat", display a 🐱 icon. For other species, show a generic 🐾 icon.

PROGRAM:

PetCard.jsx:

```
import React, { Component } from "react";
import "./PetCard.css";

class PetCard extends Component {
  constructor(props) {
    super(props);
    this.state = {
      showDetails: false
    };
  }

  toggleDetails = () => {
    this.setState({
      showDetails: !this.state.showDetails
    });
  };

  getIcon = () => {
    const species = this.props.species.toLowerCase();
    if (species === "dog") {
      return "🐶";
    }
```

```
    } else if (species === "cat") {  
      return "🐱";  
    } else {  
      return "🐾";  
    }  
  };  
  render() {  
    const {  
      name,  
      age,  
      species,  
      description  
    } = this.props;  
    return (  
      <div className="pet-card">  
        <div className="pet-icon">  
          {this.getIcon()}  
        </div>  
        <h1>{name}</h1>  
        {age < 2 && (  
          <span className="young-badge">  
            ★ Young  
          </span>  
        )}  
        <div className="pet-info">  
          <p>🐾 Species: {species}</p>  
          <p>🎂 Age: {age} years</p>  
        </div>  
        <button onClick={this.toggleDetails}>
```

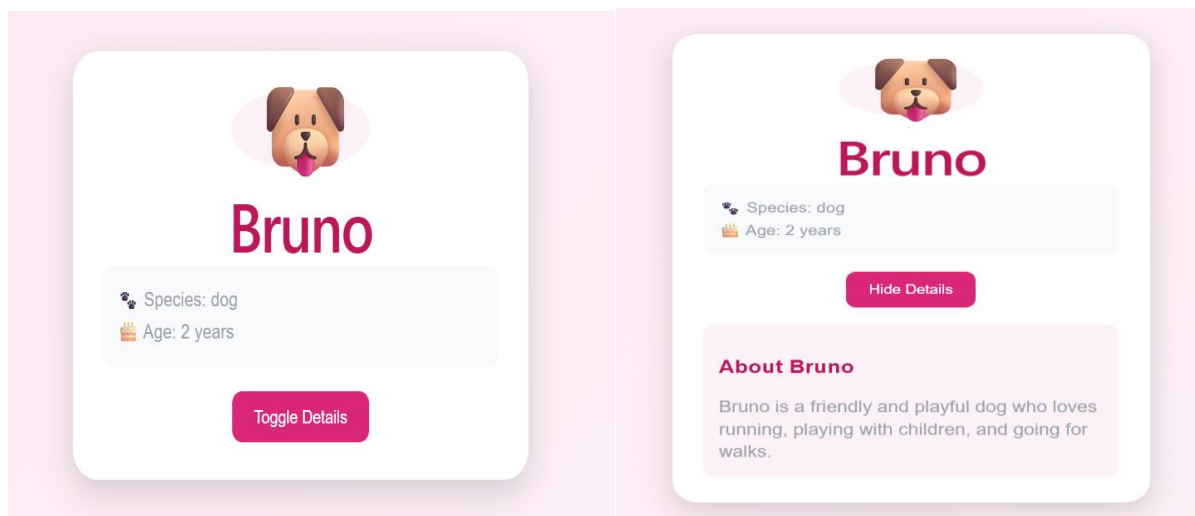
```
        {this.state.showDetails
          ? "Hide Details"
          : "Toggle Details"}
      </button>

      {this.state.showDetails && (
        <div className="description">
          <h3>About {name}</h3>
          <p>{description}</p>
        </div>
      )}
    </div>
  );
}
}
export default PetCard;

App.jsx

import PetCard from "./PetCard";

function App() {
  return (
    <PetCard
      name="Bruno"
      age={2}
      species="dog"
      description="Bruno is a friendly and playful dog who loves running, playing with
children, and going for walks."
    />
  );
}
export default App;
```

OUTPUT:**3. Real-Time Auction Platform**

Scenario: Build an auction platform where users can bid on items.

Tasks:

1. Create an AuctionItem class component that:
 - Displays itemName, currentBid, and bidderName as props.
 - Highlights the item if the currentBid exceeds \$1,000.
2. Add a "Place Bid" button that allows the user to input a bid amount (via a prompt). If the bid is higher than the currentBid, update it and show the new bidderName.
3. Display a "Hot Item" badge if the number of bids exceeds 5.

PROGRAM:**AuctionItem.jsx:**

```
import React, { Component } from "react";
import "./AuctionItem.css";
class AuctionItem extends Component {
  constructor(props) {
    super(props);
    this.state = {
      currentBid: props.currentBid,
      bidderName: props.bidderName,
      bidCount: props.bidCount
```

```
    };  
  }  
  placeBid = () => {  
    const bid = prompt(  
      `Enter your bid. Current bid: $$${this.state.currentBid}`  
    );  
    const bidder = prompt("Enter your name:");  
    if (bid === null || bidder === null) {  
      return;  
    }  
    const newBid = Number(bid);  
    if (newBid > this.state.currentBid && bidder.trim() !== "") {  
      this.setState({  
        currentBid: newBid,  
        bidderName: bidder,  
        bidCount: this.state.bidCount + 1  
      });  
      alert("Bid placed successfully!");  
    } else {  
      alert(  
        "Invalid bid! Your bid must be higher than the current bid."  
      );  
    }  
  };  
  render() {  
    const {  
      itemName  
    } = this.props;  
    const {
```



```
currentBid,
bidderName,
bidCount
} = this.state;
const hotItem = bidCount > 5;
const highBid = currentBid > 1000;
return (
  <div className={`auction-card ${highBid ? "high-bid" : ""}`}>
    <div className="auction-icon">
      
    </div>
    <h1>{itemName}</h1>
    {hotItem && (
      <span className="hot-badge">
         Hot Item
      </span>
    )}
    {highBid && (
      <span className="premium-badge">
         Premium Bid
      </span>
    )}
    <div className="bid-section">
      <p>Current Bid</p>
      <h2>
        ${currentBid}
      </h2>
      <p>
         Highest Bidder: <b>{bidderName}</b>
      </p>
    </div>
  </div>
)
```

```
    </p>
    <p>
       Number of Bids: {bidCount}
    </p>
  </div>
  <button onClick={this.placeBid}>
     Place Bid
  </button>
</div>

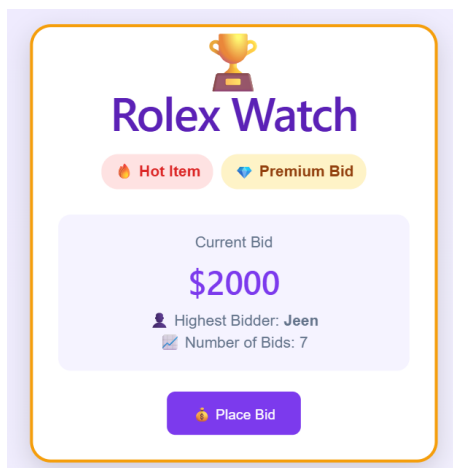
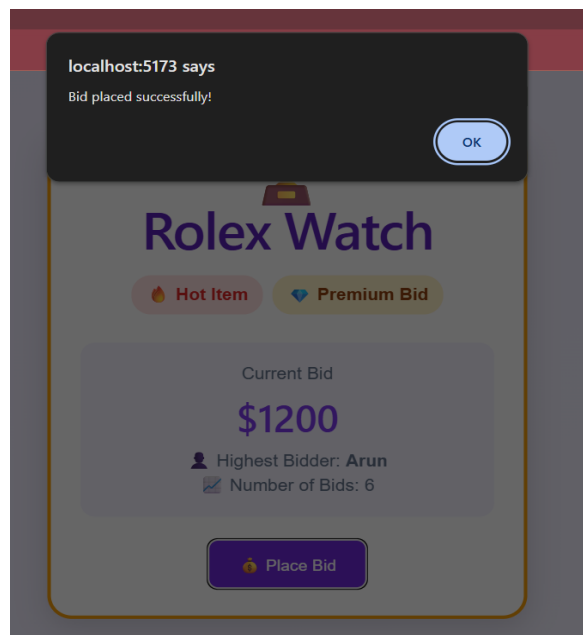
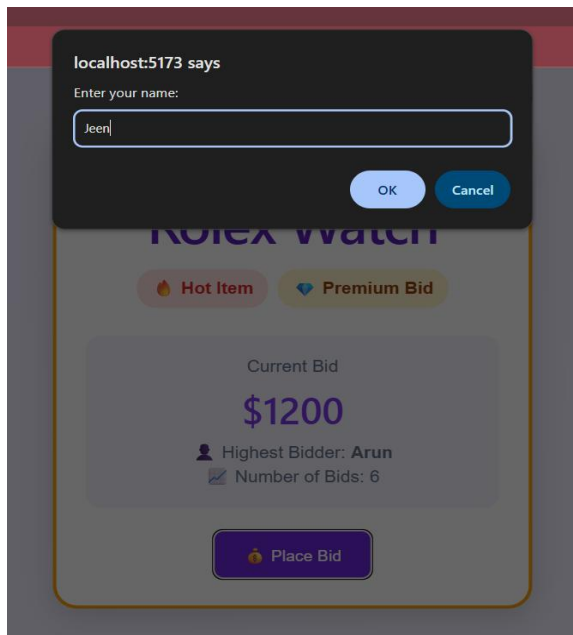
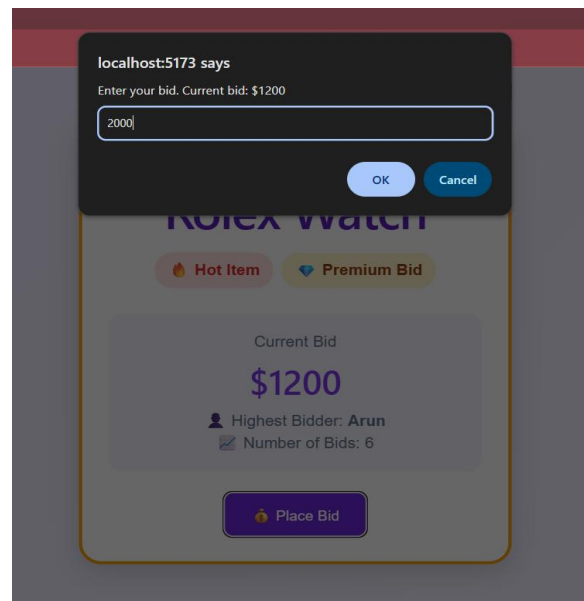
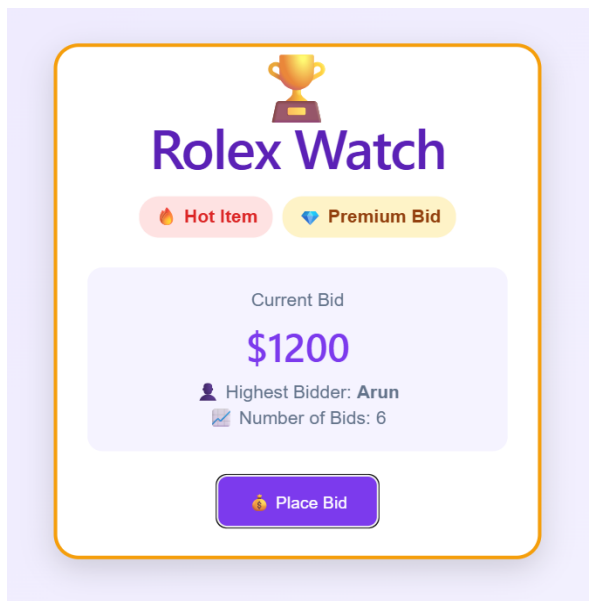
);
}
}

export default AuctionItem;

App.jsx:
import AuctionItem from "./AuctionItem";

function App() {
  return (
    <AuctionItem
      itemName="Rolex Watch"
      currentBid={1200}
      bidderName="Arun"
      bidCount={6}
    />
  );
}

export default App;
```

OUTPUT:

Problem Understanding & Approach (10)	Implementation & Execution (15)	Tool/Technology Usage (10)	Code Quality & Documentation (5)	Viva (5)	Time Management (5)	Total (50)

Result:

Thus, the Program on React using Function Component is completed and the output is verified.